

### Screen 1: Look Up Classes

|                                  |                                |                               |
|----------------------------------|--------------------------------|-------------------------------|
| Fall 2024                        |                                |                               |
| Department: Computer Engineering |                                |                               |
| 1000                             | Personal Computer Applications | <a href="#">View Sections</a> |
| 1200                             | Intro to Computers             | <a href="#">View Sections</a> |
| 1210                             | Fundamentals of Computing I    | <a href="#">View Sections</a> |

### Screen 2: Sections

| Dept | Course | Section | Credit | Title                       | Days | Time        | Capacity | Actual | Rem | Instructor   |
|------|--------|---------|--------|-----------------------------|------|-------------|----------|--------|-----|--------------|
| COMP | 1210   | 1       | 3      | Fundamentals of Computing I | MW   | 12:00-13:15 | 25       | 25     | 0   | Joe Anne     |
| COMP | 1210   | 2       | 3      | Fundamentals of Computing I | MW   | 13:30-14:45 | 25       | 25     | 0   | Joe Anne     |
| COMP | 1210   | 3       | 3      | Fundamentals of Computing I | MW   | 15:00-16:15 | 25       | 21     | 4   | Jenny Tutone |
| COMP | 1210   | 4       | 3      | Fundamentals of Computing I | MW   | 16:30-17:45 | 25       | 13     | 12  | Jenny Tutone |

### Screen 3: Grade Summary for Spring 2022 for Student Nusret Gökçe (Student number 8675309)

| Dept         | Course | Section   | Title                       | Grade      | Credit | Earned | GPA points |
|--------------|--------|-----------|-----------------------------|------------|--------|--------|------------|
| COMP         | 1200   | 3         | Intro to Computers          | AA         | 3      | 3      | 12         |
| COMP         | 1210   | 2         | Fundamentals of Computing I | DD         | 3      | 3      | 3          |
| MATH         | 1430   | 1         | Calculus I                  | BB         | 3      | 3      | 9          |
| PHYS         | 1200   | 1         | Physics for Computer Majors | FF         | 4      | 0      | 0          |
| GPA Summary  |        |           |                             |            |        |        |            |
|              |        | Attempted | Earned                      | GPA Points | GPA    |        |            |
| Current Term |        | 13        | 9                           | 24         | 1.85   |        |            |
| Overall      |        | 80        | 71                          | 184        | 2.30   |        |            |

1. "Assume the first thing we did was bring the source data into a single unnormalized table. How many columns would be in this table?

Name the columns. For example, if you think there should be two columns your response should be in the following format:

2

Name

Email address

2. There are at least two (2) columns that could be found from the source data that we would call "derived data". Choose two of these columns and for each explain why it should be considered derived (i.e. what is the underlying source data) and how you would choose to represent this data in your database. Your response should be the

following format:

1- Name

1- It is derived because it is my grandfather's name

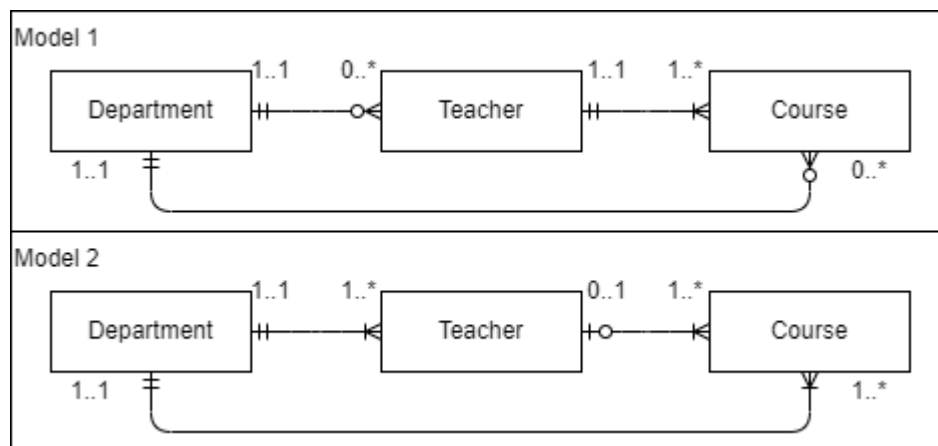
2- I would represent this data by ...

2- Email address

1- It is derived because it comes from my favorite star wars character and the domain of my email provider

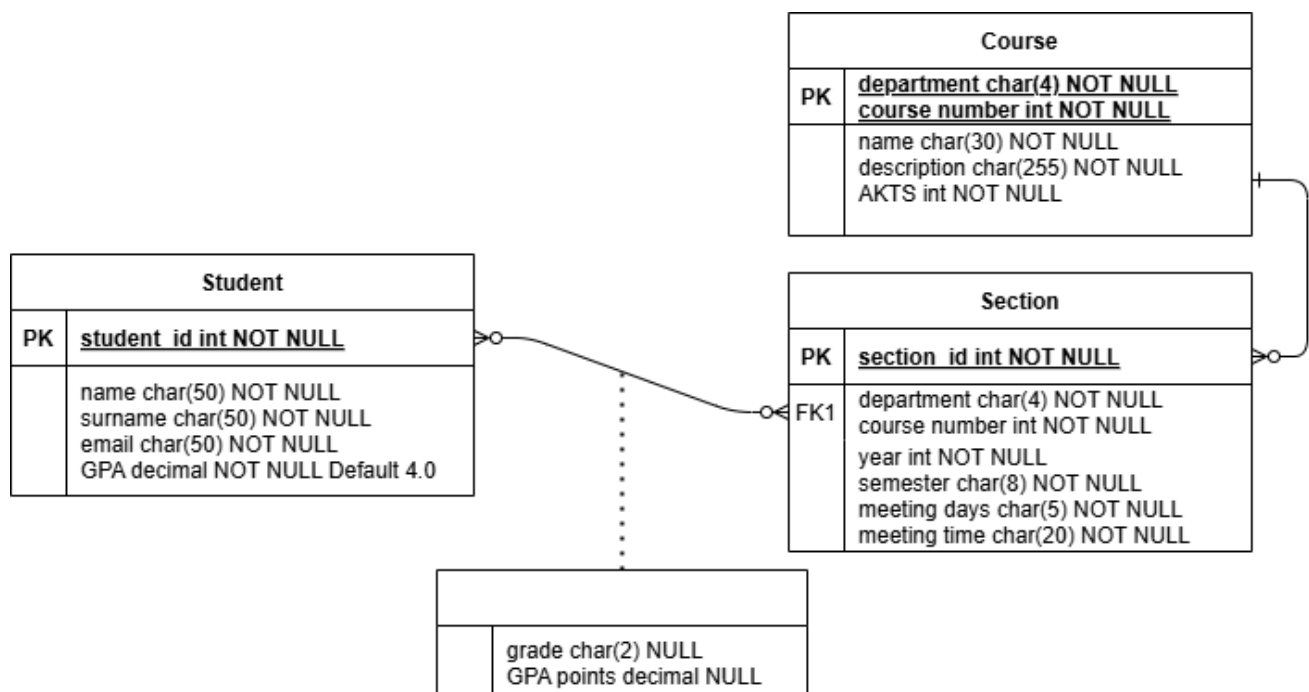
2- I would represent this data by ...

3. Use SQL to create the tables that you would have in your 3NF version of this DB. At minimum, you should have the following tables: Course Section Student Be sure to include necessary constraints as part of your SQL statements.



4. Model 1 design flaw:

5. Model 2 design flaw:



6. Identify all students (number, name and surname) that are taking COMP 1210 in the Fall 2025.

7. List all courses and sections from the MATH department along with the number of students who took each section.
8. Add a new course to the system for the COMP department. You choose the ID, title, and credit for the course.
9. A student with your student number wants to drop COMP 1210 for Fall 2025. Please remove this student from the section he/she is in.
10. Add a new column for storing in which room a course section will meet.
11. All COMP 1210 course sections will meet in room 102. Store this information in the appropriate table.
12. A student named Leonardo DiCaprio (Student number 1458914) has earned a grade of 'BA' in his COMP 3000 course. Please update his grade and make sure his GPA is updated as well.
13. Give privileges to view, add, and change data on all tables to the role REKTOR. She should not be able to remove data.
14. Create a view for the rector to see a list of all students and their overall GPA. Sort the students by highest GPA first.
15. One of the above tasks would have been easier to code if a certain column in the DB had been designed differently. However, one of the above tasks would have been more difficult. Answer the following: 1- Which column could have been designed differently and how? 2-Which task would have been easier? 3-Why? Which task would have been harder? Why?
16. How many relations will be created to represent this ER Model?
17. Using only the letter identifier for each entity, provide appropriate names for the relations mapped from the ER model. In other words, if your solution would make A a table, "A" should be part of your answer. If you think A and B would combine to make a single table "AB" should be part of the answer. Cevap: nedir
18. How many foreign keys will be in the relation representing X?
19. Which relation(s) will have no FK columns?
20. If the relationships were all changed to 1:1 with mandatory participation (1..1 on all), how many relations would be made from the modified version of the ER Model?
21. If the relationships were all changed to : with optional participation (0..\* on all), how many relations would be made from the modified version of the ER Model?